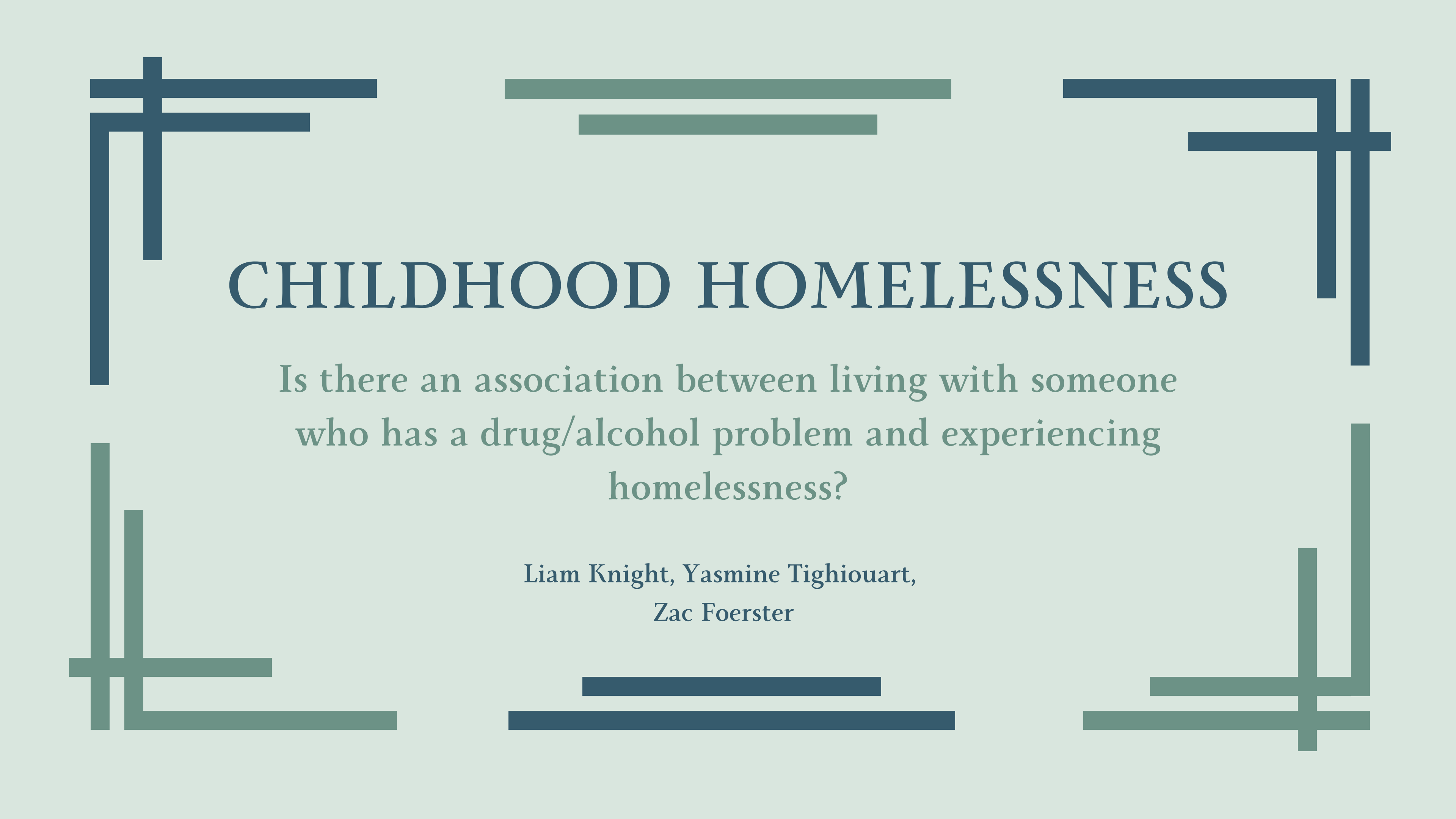




CHILDHOOD HOMELESSNESS

Is there an association between living with someone
who has a drug/alcohol problem and experiencing
homelessness?

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RESEARCH QUESTION

Hypothesis



- For the children in this study, there will be a positive association between living with someone with a drug or alcohol problem and ever having experienced homelessness.
 - When compared to children who have not lived with drug/alcohol problems in the home

Covariates



- This association will vary based on race, age, and sex.

*Note on terminology: study used term “homeless”, we recognize shift to “unhoused”

STUDY DESIGN AND DATA

Cross-Sectional

- The data was collected at one point in time. We cannot infer causality because this is a cross-sectional study, but we can state an association

Population Sampled

- 385,000 addresses sampled from US Census data and sent questionnaire by mail
- Subsampling stratified by ages to increase representation of young children
- 53,699 children in our study, ages 0-17

Data Collection

- Collected data from the 2023 National Survey of Children's Health (NSCH) database

Missing Data

- There were a few missing/NA values, but we had enough data to include it without affecting our analysis

MORE ON DATA

Exposure/Outcome Variables

- EVERHOMELESS
 - (“...ever been homeless or lived in a shelter?”)
- ACE₉
 - (“...ever experienced living with anyone who had a problem with alcohol or drugs?”)

Covariates

- Age, sex, and race (White, Black or African American, American Indian or Alaska Native, Asian, Native Hawaiian and Other Pacific Islander, Two or More Races)

Inclusion/Exclusion

- Note no age exclusion

Logistic Regression Modeling

- Bivariate logistic regression
- Adjusted model with covariates

Variable	N	N = 55,162 ¹
EverHomeless	53,699	
No		52,572 (98%)
Yes		889 (1.7%)
Unknown		238 (0.4%)
Missing Data		1,463
Drug_Alc_Prob	52,952	
Has Not		48,617 (92%)
Has		4,335 (8.2%)
Missing Data		2,210
Sex	55,162	
Male		28,330 (51%)
Female		26,832 (49%)
Age	55,162	
Mean (SD)		8.3 (5.3)
Median (Q1-Q3)		8.0 (4.0-13.0)
Min-Max		0.0-17.0
Race	55,162	
White		42,095 (76%)
Black		3,686 (6.7%)
Native American		507 (0.9%)
Asian		3,632 (6.6%)
Pacific Islander		260 (0.5%)
Two or More Races		4,982 (9.0%)

¹n (%)

TABLE ONE: DESCRIPTIVE STATISTICS

- 1.7% Ever homeless
- 8.2% Ever lived with someone with a drug/alcohol problem
- Even split amongst genders
- Median age of 8 - right in the middle of range
- Low counts of Native Americans + Pacific Islanders - possibly an issue if too small
- Missing data (EVERHOMELESS + DRUG_ALC_PROB)

TABLE TWO: TABULAR ANALYSIS

Has Ever Been Homeless					
Drug__Alc__Prob	No	Yes	Unknown	NA__	Total
Has Not	47,901 (98.5%)	433 (0.9%)	87 (0.2%)	196 (0.4%)	48,617 (100.0%)
Has	3,762 (86.8%)	410 (9.5%)	138 (3.2%)	25 (0.6%)	4,335 (100.0%)
NA	909 (41.1%)	46 (2.1%)	13 (0.6%)	1,242 (56.2%)	2,210 (100.0%)
Total	52,572 (95.3%)	889 (1.6%)	238 (0.4%)	1,463 (2.7%)	55,162 (100.0%)

Note:

- Relatively low number of missing values
- Relatively low number of “unknown” answers to homeless question, though much higher for those who have the exposure

TABLE THREE: BIVARIATE ANALYSIS

- Much higher rates of homelessness in those who have lived with someone with a drug/ alcohol problem
- Even split of gender amongst both subgroups (slightly more males in homeless group)
- Higher mean/median age in homeless group
- Small but usable amounts of Native Americans + Pacific Islanders in homeless subgroup
- Missing data - not an issue

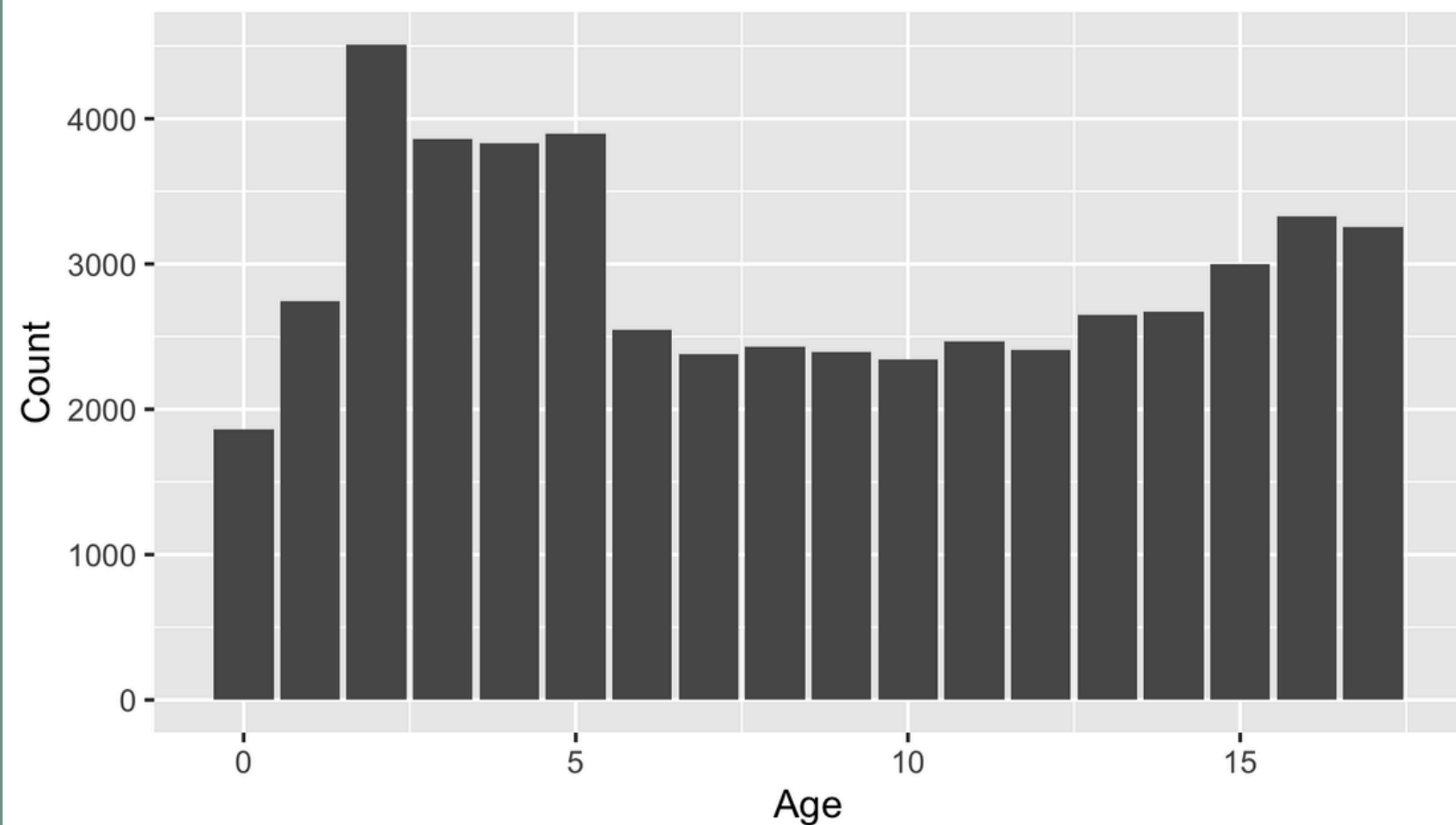
Variable	N	Ever Been Homeless			p-value ²
		No N = 52,572 ¹	Yes N = 889 ¹	Unknown N = 238 ¹	
Drug_Alc_Prob	52,731				<0.001
Has Not		47,901 (93%)	433 (51%)	87 (39%)	
Has		3,762 (7.3%)	410 (49%)	138 (61%)	
Missing Data		909	46	13	
Sex	53,699				0.006
Male		26,953 (51%)	504 (57%)	121 (51%)	
Female		25,619 (49%)	385 (43%)	117 (49%)	
Age	53,699				<0.001
Mean (SD)		8.3 (5.3)	11.1 (4.6)	10.5 (4.8)	
Median (Q1-Q3)		8.0 (4.0-13.0)	12.0 (7.0-15.0)	11.0 (7.0-15.0)	
Min-Max		0.0-17.0	0.0-17.0	0.0-17.0	
Race	53,699				
White		40,416 (77%)	518 (58%)	145 (61%)	
Black		3,304 (6.3%)	172 (19%)	38 (16%)	
Native American		451 (0.9%)	34 (3.8%)	12 (5.0%)	
Asian		3,486 (6.6%)	18 (2.0%)	17 (7.1%)	
Pacific Islander		224 (0.4%)	12 (1.3%)	3 (1.3%)	
Two or More Races		4,691 (8.9%)	135 (15%)	23 (9.7%)	

¹n (%)

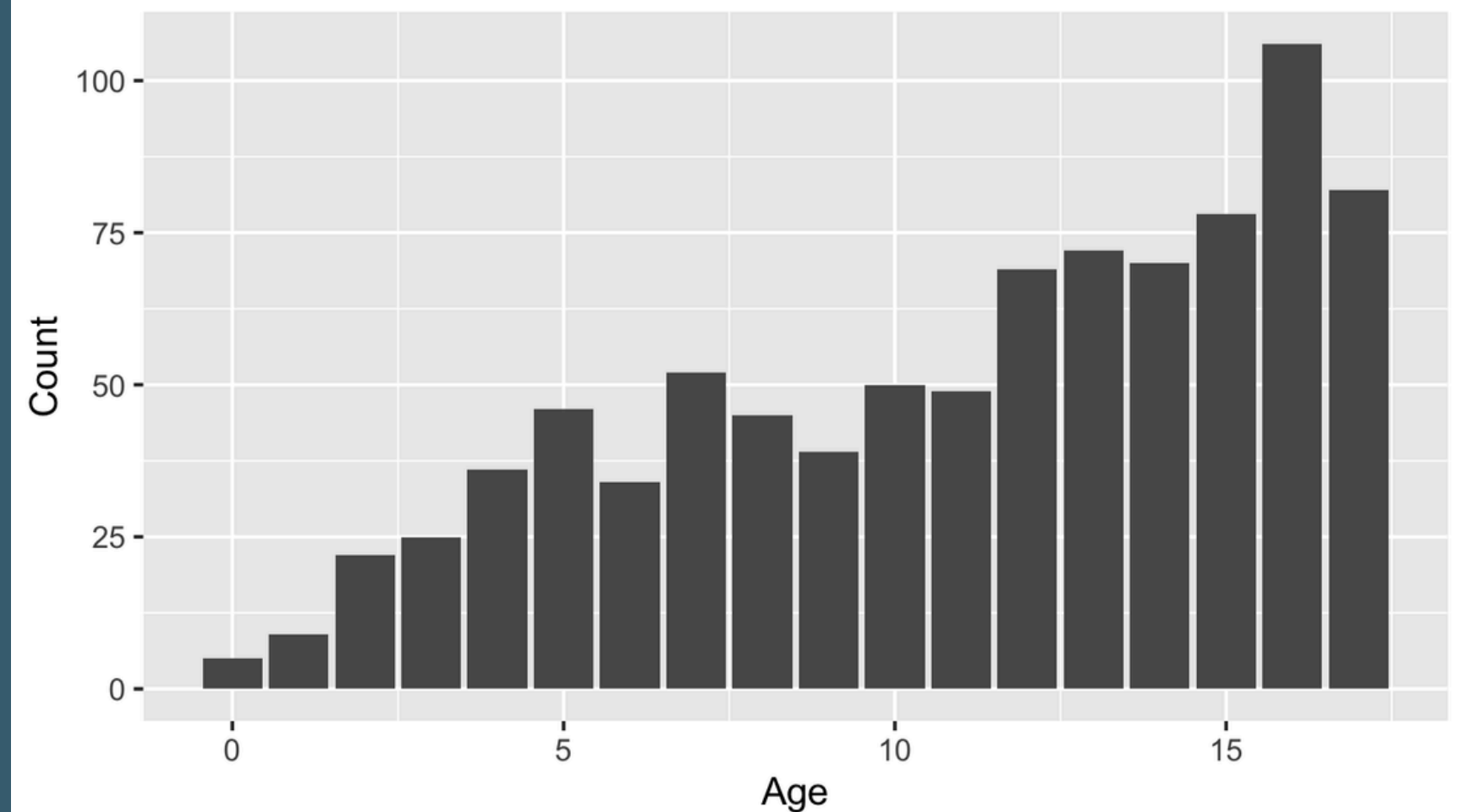
²Pearson’s Chi-squared test; Kruskal-Wallis rank sum test; NA

DATA VISUALIZATION: AGE BETWEEN GROUPS

Histogram of Age (Not Homeless)



Histogram of Age (Homeless)



- Much more uniform distribution in non-homeless group (slightly younger distribution)
- Homeless group appears to increase relatively linearly with age (even after attempting to select more young children)

MODEL ONE: UNADJUSTED

Drug_Alc_Prob	estimate	std.error	statistic	p.value	s.value	conf.low	conf.high	df
Has Not	0.0107391	0.0004684	22.92759	0	384.0415	0.0098211	0.0116572	Inf
Has	0.1271462	0.0050744	25.05636	0	457.8525	0.1172005	0.1370918	Inf

*Estimates are given in probabilities

- People who have not lived with someone with a drug/alcohol problem have a 1% probability of ever being homeless.
- People who have lived with someone with a drug/alcohol problem have a 12.7% probability of ever being homeless.

P-values are significant; confidence intervals show very little possible variation.

MODEL TWO: ADJUSTED

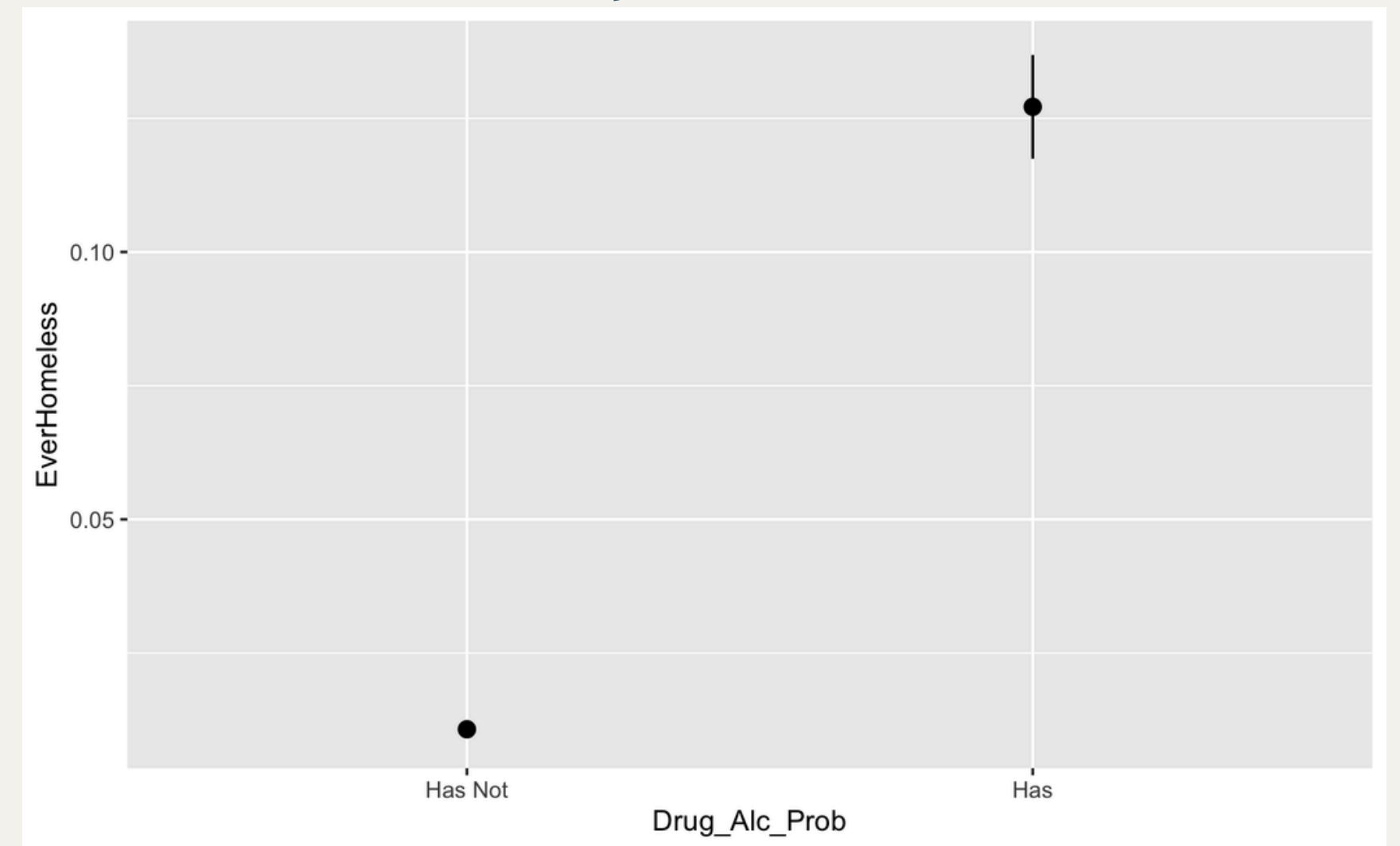
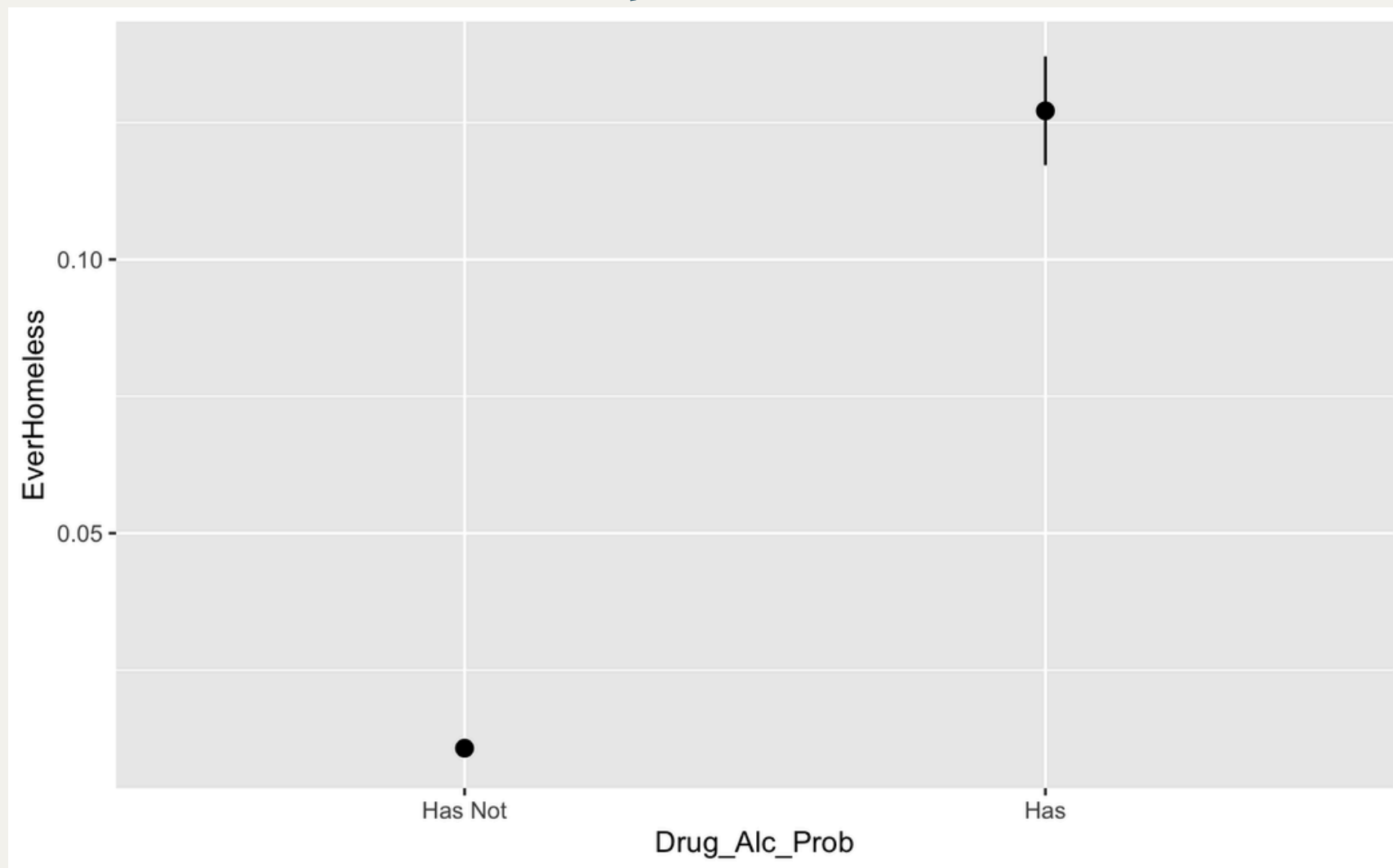
term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	0.01	0.09	-59.52	0.00	0.00	0.01
Drug_Alc_ProbHas	11.96	0.07	36.77	0.00	10.48	13.65
Age	1.06	0.01	8.70	0.00	1.05	1.07
SexFemale	0.85	0.06	-2.52	0.01	0.75	0.96
RaceBlack	4.73	0.09	17.43	0.00	3.96	5.62
RaceNative American	3.85	0.18	7.64	0.00	2.70	5.39
RaceAsian	0.99	0.18	-0.04	0.97	0.68	1.40
RacePacific Islander	4.02	0.31	4.54	0.00	2.10	7.06
RaceTwo or More Races	2.08	0.10	7.53	0.00	1.71	2.51

*Estimates are given in odds

- All (covariate) factors have reasonably small effect sizes.
- Age (very slightly) increases risk of homelessness - adds up.
- Male individuals have a higher likelihood of experiencing homelessness.
- All races other than White (and Asian) increase likelihood of experiencing homelessness.

DATA VISUALIZATION:

UNADJUSTED VS. ADJUSTED



The two models are very similar

```
Start:  AIC=8649.74
EverHomeless ~ Drug_Alc_Prob + Age + Sex + Race
```

	Df	Deviance	AIC
<none>		8631.7	8649.7
- Sex	1	8638.1	8654.1
- Age	1	8709.4	8725.4
- Race	5	8948.1	8956.1
- Drug_Alc_Prob	1	9845.3	9861.3

```
Call:  glm(formula = EverHomeless ~ Drug_Alc_Prob + Age + Sex + Race,
  family = binomial(link = "logit"), data = dat)
```

Coefficients:

(Intercept)	Drug_Alc_ProbHas	Age
-5.273378	2.481524	0.057365
SexFemale	RaceBlack	RaceNative American
-0.162963	1.553791	1.348348
RaceAsian	RacePacific Islander	RaceTwo or More Races
-0.006418	1.391054	0.731884

```
Degrees of Freedom: 52730 Total (i.e. Null);  52722 Residual
(2431 observations deleted due to missingness)
```

```
Null Deviance:      10440
```

```
Residual Deviance: 8632      AIC: 8650
```

FINDING THE MOST PARSIMONIOUS MODEL:

- Backwards Selection AIC Minimization
- Already the most parsimonious model
- Tested with both adjusted model and a model with missing values removed



STRENGTHS



Large Sample Sizes

- Children with data on homelessness: 53,699
- Children who have been homeless: 889
- Children who have not been homeless: 52,572

Many variables: easier to adjust for covariates

- We looked at age, sex, and race, but possible to adjust for many more covariates for further study

Hypothesis Generation

- Strong association between living with someone with a drug/alcohol problem and having ever been homeless, but also...
 - Race appears to have strong associations
- 



LIMITATIONS



Data Collection

- Self-reported data
- Missing data
- Language barriers in mailings

Cross-Sectional Study Design

- Cannot determine causality
- Cannot determine temporality between exposure/outcome variables

Recall/Information Bias

- Self-reported data -> inaccurate reporting
 - Non-response bias
- 

CONCLUSIONS

- There is an association between a child having lived with someone with a drug/alcohol problem and having experienced homelessness.
- This association has a clinically significant effect size; however, the covariates do not affect the model sufficiently to be considered majorly impactful.
- Future models should attempt to adjust for more covariates and include these, as they are included in the most parsimonious model.